

Meghan Gillen
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Assignment 1 - EE 446

Problem 1e:

So far, we have tried quantization, knowledge distillation and pruning. Within quantization there were three types: int8, float16 and dynamic quantization. The baseline model achieved an accuracy of 88.8%. The file size was 14.06 KB. For the rest of the models..

Model	Accuracy	Size
Int8 Quantization:	41%	5.73 KB
Float16 Quantization:	89%	8.94 KB
Dynamic Quantization:	89%	8.16 KB
Pruned Model:	61%	14.09 KB
Knowledge Distillation Student Model:	39%	6.11 KB

The best method from all of these seems to be dynamic quantization. After that, it was the Float16 quantization. These are the best because they completely maintain the original accuracy of the base model with 41.9% and 36.4% size reduction. Int8 quantization and knowledge distillation reduce the size of the model much more, but have huge reductions in accuracy and would not be practically useful, since the model would not work even if it could be deployed on smaller systems.

I think combining multiple methods would reduce accuracy beyond what is useful in size reduction but one method that could be tested is instead making a more simple/smaller model. I used the structure of the student model and trained it on the original dataset. Then, I quantized it to Float16. From this, my results were an accuracy of 98% and a size of 5.02 KB. This is a huge improvement on anything before. Since the dataset is simple, this makes sense as the larger models might have been unnecessarily complicated and it is important in ML to make the models as simple as possible (if the data is simple, the model should be as well).

To further investigate the size, I checked the size of the model before it was quantized and it was 6.12 KB. So, the size reduction of the quantization was not huge, but starting with a smaller model let the quantized model be smaller than any size reduction from the model's above as well as more accurate.

Problem 2a:

The learning blocks that are available are Classification, Regression, Anomaly Detection (GMM), Anomaly Detection (K-means) and Anomaly Detection.

Classification: Assigns an input to a category. Ex. Checking if a disease is present in an image or not.

Regression: Finding an exact value based on input. Ex. Finding the exact waiting time for a waiting room depending on the patient's condition.

Anomaly Detection (GMM): Find unlikely events based on comparison to the normal distribution. Ex. Detecting abnormal sensor readings in industrial equipment

Anomaly Detection (K-means): Finds unlikely events based on distance to clusters. Ex. Identifying unusual network traffic patterns

Anomaly Detection: Finds unlikely events based on deviation from other inputs. Ex. Checking for fraud in a bank.

Problem 2b:

All layers under Classifier are: Dense, 1D Convolution / pooling, 2D Convolution / pooling, Reshape, Flatten, and Dropout.

Dense is usually used for processed data, it's a simple layer. The 1D Convolution / pooling is used for spatial information along a singular dimension. The 2D is the same but with 2 dimensions. These use raw data. The reshape is used to turn 1D data into multi dimensional data, while Flatten does the opposite. These are used on the input and output to a convolutional layer, respectively. Finally there is dropout which cuts a couple of the connections to reduce overfitting.

Problem 2c:

You can only quantize to int8 or keep the original Float32.

Problem 2d:

The options for input modalities are image and speech generation. There are two image options which are.. Black Forest Labs Flux-Pro Synthetic Image Generator and DALL-E 3 Synthetic Image Generator. The speech one is Whisper Synthetic Voice Generator.

ML needs lots of data which is not always easy to get. As well there are specific requirements for the data which help make models better. By using synthetic data you can help balance datasets and improve model robustness by making the data more diverse.